Drug Interaction Severity Prediction

Deep-Pharma Multimodal Intelligence — DataHack 3.0

Aidan Colvin | BIOL/BIOS Graduate Project | April 2026

Generative AI provided limited assistance in line with the course policy on usage of AI.

15,284	126	3	0.5134
Drug Pairs	Features	Tasks	Kaggle Score

UNC Chapel Hill

Biomedical & Health Informatics

The Problem

Clinical Context

- 1 in 4 U.S. adults filled 3+ prescriptions monthly (2015–2018)
- Over 2/3 of adults aged 65+ take 3 or more drugs simultaneously
- This practice — polypharmacy — creates compounding DDI risk
- Clinical trials cannot cover all possible drug combinations
- Pharmacovigilance systems detect harmful interactions reactively — often after patient harm
- Federal DDI safety guidance has been revised since the late 1990s due to documented failures

Research Questions

RQ1 — Supervised Prediction

Can a multi-task ML model accurately predict DDI severity, adverse event occurrence, and PRR risk from pharmacological and molecular drug pair features?

RQ2 — Unsupervised Exploration

What natural groupings exist within drug pair interaction profiles, and do these clusters align with meaningful differences in severity and side effect patterns?

(Extension beyond competition scope)

Goal: A scalable screening layer to strengthen clinical decision support at the point of prescribing

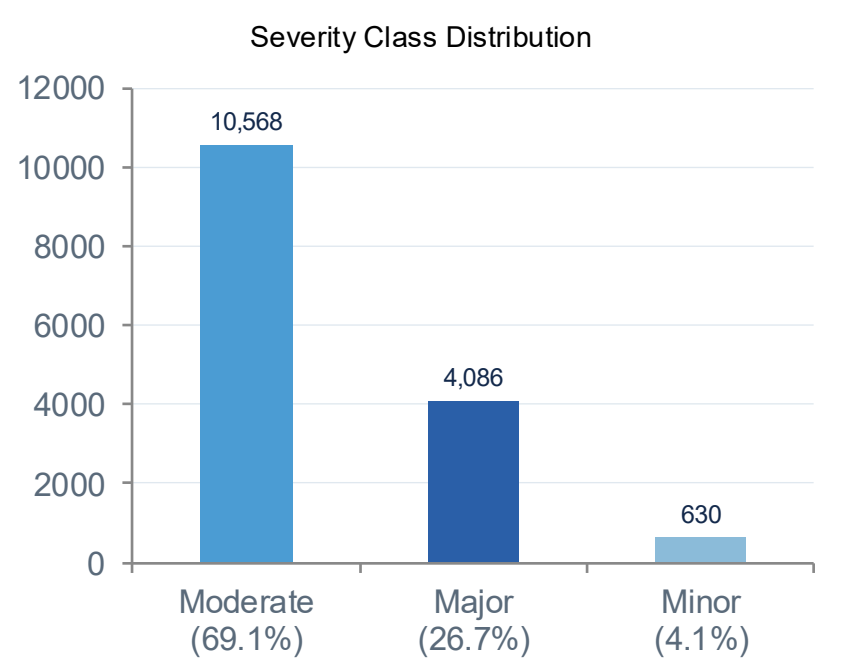
Dataset Overview

15,284
Drug Pair Obs.

126
Feature Columns

50
Adverse Events

16.8:1
Class Imbalance



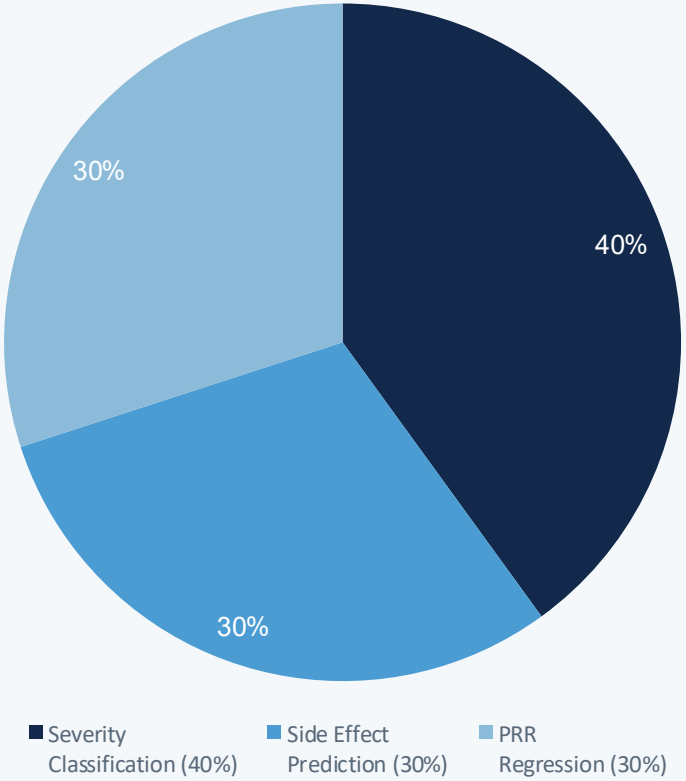
Feature Missingness

Feature	Missing %
CYP450_Enzymes_A	59.5%
CYP450_Enzymes_B	47.1%
Warning_A	31.0%
Indication_A	19.1%
Warning_B / Indication_B	11.3%

Adverse Event & PRR Distribution

- 68.5% of drug pairs have zero adverse events reported
- 14.7% of pairs have 40 or more adverse events (bimodal)
- Only 21.5% of PRR entries are non-zero
- Non-zero PRR: mean 3.89, median 2.50, max 240.0

Evaluation: Hardcore Clinical Score



Task 1: Severity Classification

Weight: 40% | Metric: Macro F1

3-class: Major / Moderate / Minor
Gradient Boosting — final model

Score

0.5666

Task 2: Side Effect Prediction

Weight: 30% | Metric: Micro F1

50-label binary prediction
MultiOutput Logistic Regression

Score

0.5772

Task 3: PRR Regression

Weight: 30% | Metric: Inv. RMSE

50-label continuous risk score
Ridge Regression (masked eval)

Score

0.2023

Estimated Composite: 0.4605 | Kaggle Submitted Score: 0.5134

Methodology

1

Raw Text Concat

Mechanism, pharmacodynamics, ADME, CYP450, indications, warnings, toxicity → single string per drug pair

2

Stratified Split

80/20 train/test split stratified on severity class to maintain class proportions

3

TF-IDF Vectorization

2,000 tokens, English stop word removal. Fit on training split only — no data leakage

4

Shared Feature Matrix

All three tasks use the same TF-IDF representation. SMILES included as raw text tokens

Models by Task

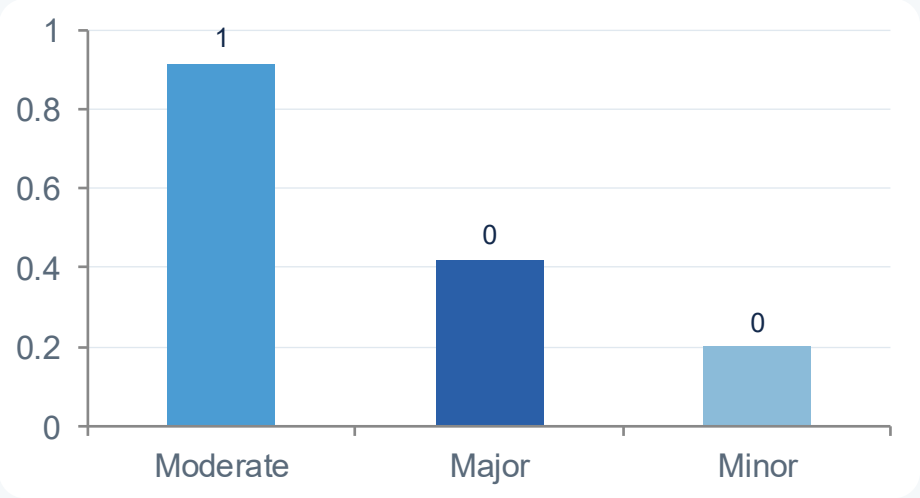
Severity Classification	Logistic Regression · Random Forest · Gradient Boosting (✓ selected)	*GB: n_estimators=100, max_depth=3, lr=0.1*
Side Effect Prediction (50 labels)	MultiOutputClassifier wrapping Logistic Regression — fit independently per label	*max_iter=1000*
PRR Regression (50 labels)	Multi-output Ridge Regression — predictions clipped at zero	*Alpha (default); sparse-stable*
Unsupervised Clustering	K-means on 50 binary AE columns — silhouette sweep k=2 to 6	*Random state fixed for reproducibility*

Results: Severity Classification

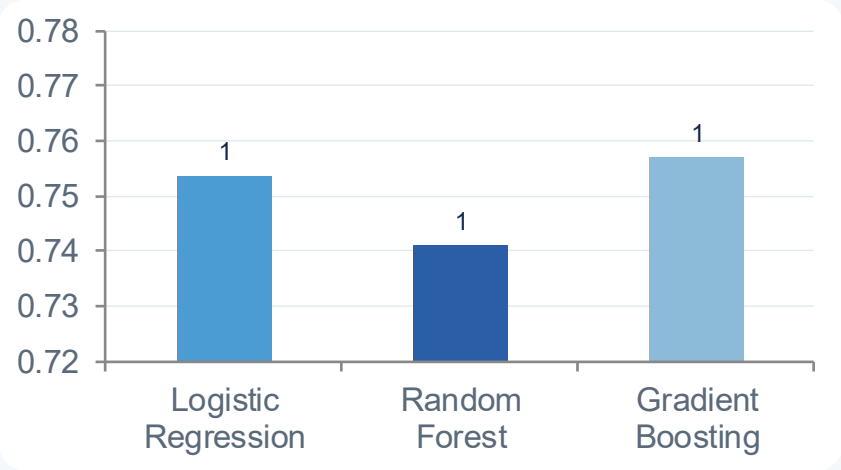
Model Comparison — Test Set

Model	Accuracy	Macro F1	AUC
Logistic Regression	0.7694	0.5395	0.9241
Random Forest	0.7661	0.5639	0.9146
Gradient Boosting ✓	0.7746	0.5666	0.9252

Per-Class Recall — Gradient Boosting



5-Fold Cross-Validation (Training Subset)



All models: AUC > 0.91. Strong class-level discrimination across unseen scaffolds.

AUC delta < 0.008 across all models — further gains require feature engineering, not model switching.

Minority class (Minor) recall only 19.8%. 16.8:1 imbalance drives this failure. Major recall 42–50%.

Results: Side Effects & PRR Regression

Binary Side Effect Prediction (50 labels)

Metric	Value
Micro F1	0.5772
Macro F1	0.5698
Mean per-label AUC	0.8938

Per-label F1 range

0.666	0.662	0.570	0.499
Dyspnoea (best)	Nausea	Mean	Cond. Aggravated (worst)

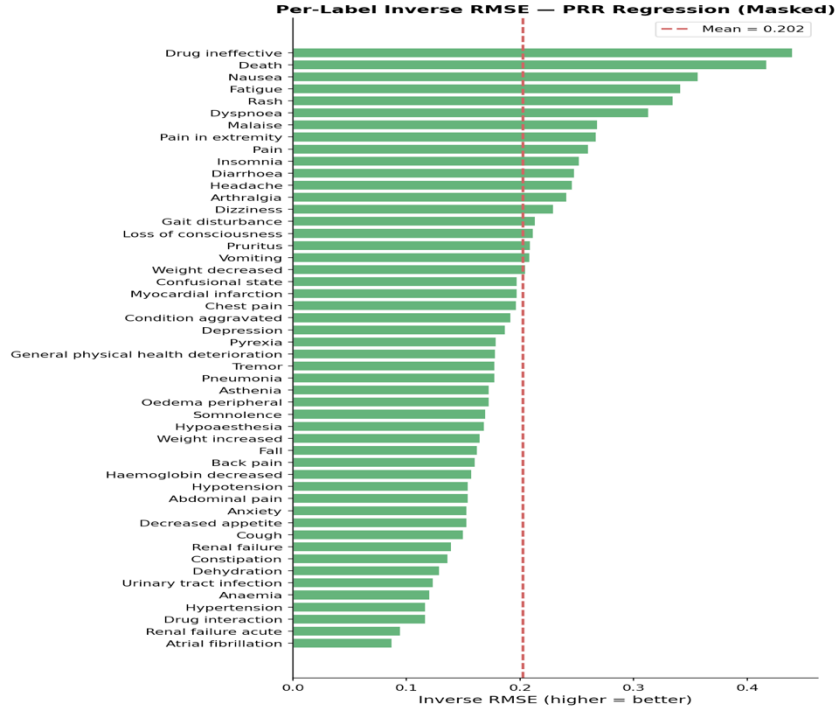
High-prevalence systemic events (Dyspnoea, Nausea, Fatigue) predicted most reliably. Rare events (MI, Loss of Consciousness) show weakest F1. Mean AUC 0.89 confirms strong ranking ability.

Interpretation

TF-IDF token features carry pharmacological signal sufficient for binary classification (AUC 0.89+) but lack structural resolution for continuous PRR quantification. SMILES were treated as raw text tokens — true molecular fingerprints would improve PRR substantially.

PRR Regression (50 labels, Masked Eval)

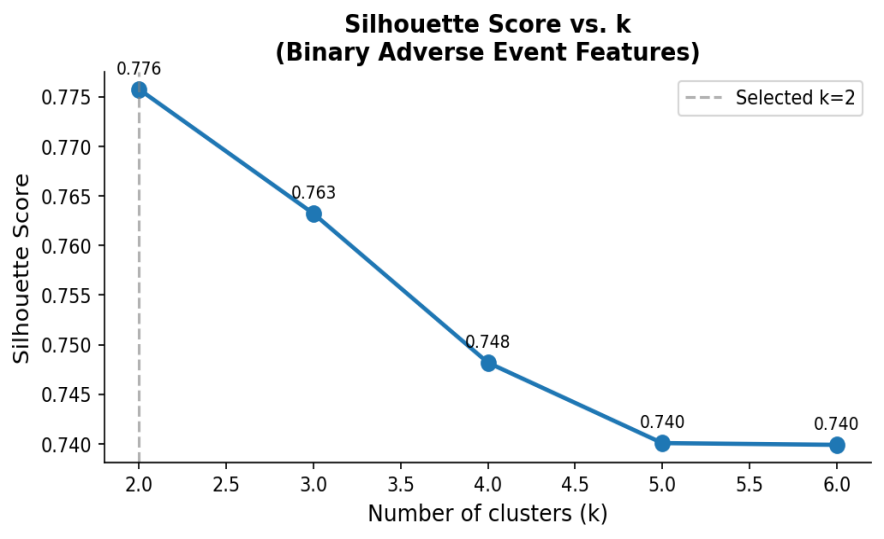
Metric	Value
Mean Masked RMSE	4.576
Mean Inverse RMSE	0.2022

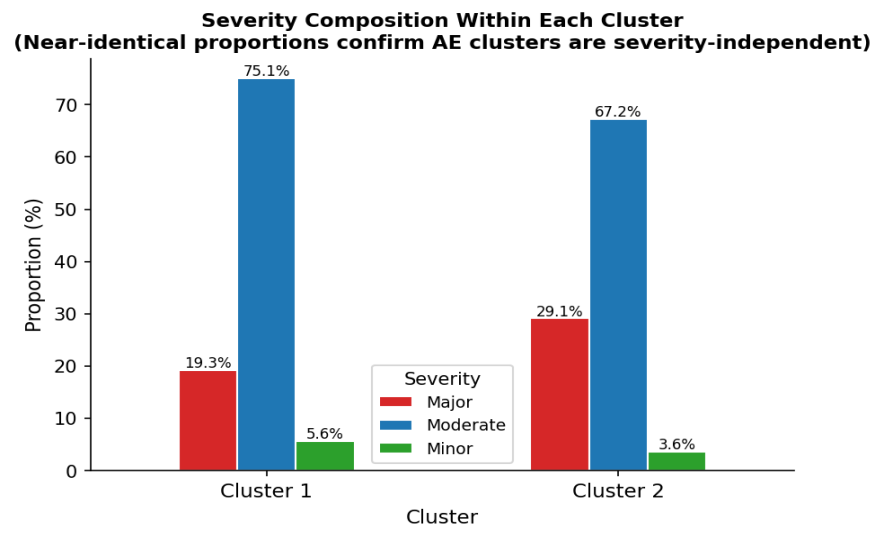


RQ2: Unsupervised Clustering

How Were the Clusters Determined?

K-means was applied to the 50 binary adverse event columns only (not severity or text features). K was selected by sweeping k=2 through k=6 and choosing the value with the highest silhouette score — a measure of how similar each point is to its own cluster vs. other clusters (higher = better). k=2 scored 0.776 and declined monotonically to 0.740 at k=6.

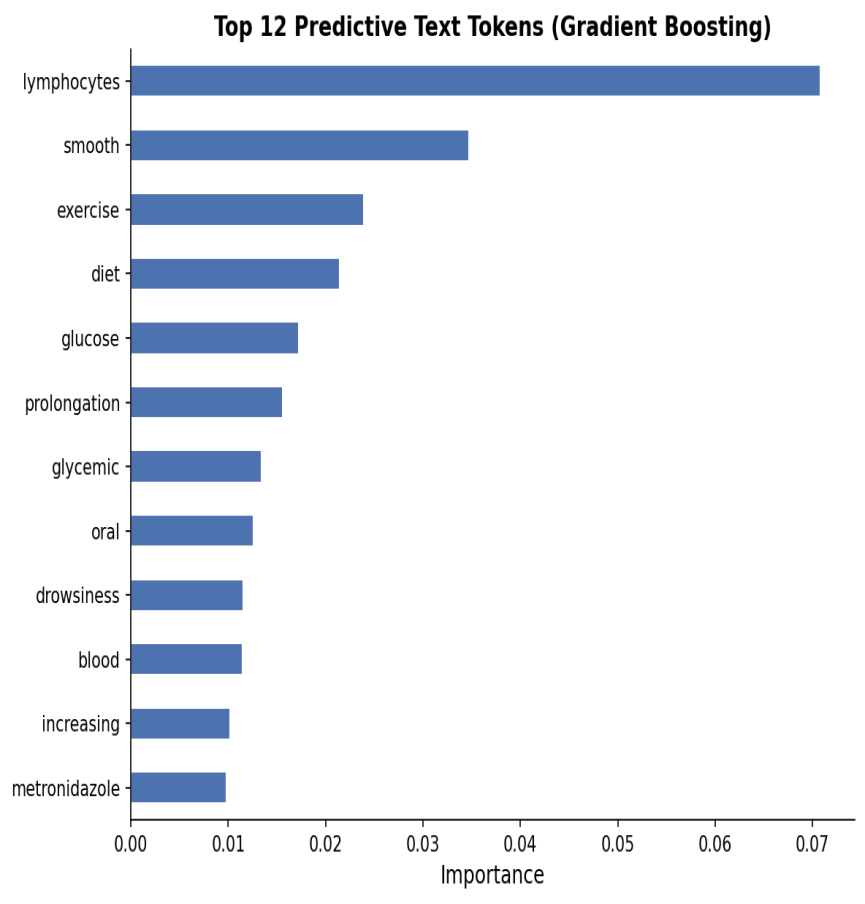




Key Finding (RQ2 — Meaningful Negative Result)

Cluster 1: high AE burden (prevalence 0.84–0.97). Cluster 2: near-zero (0.02–0.05). But severity composition is near-identical: Moderate = 75.1% vs 67.2%; Major = 19.3% vs 29.1%. Adverse event burden and interaction severity are independent dimensions.

Feature Importance & Pharmacological Insights



Pharmacological Interpretation

"lymphocytes" (MDI: 0.07)

Immunosuppressants (e.g., tacrolimus, cyclosporine) heavily use this term — high DDI risk class

"glucose / glycemic / diet" (MDI: 0.017–0.022)

Diabetic medications alter glucose metabolism; DDIs here can cause dangerous hypoglycemia

"prolongation" (MDI: 0.016)

QT prolongation — cardiac arrhythmia risk flagged across many high-alert drug classes

"smooth" (MDI: 0.034)

Smooth muscle effects linked to calcium channel blockers and antispasmodics

"drowsiness / oral" (MDI: ~0.011)

CNS depressants + route of administration markers

Additional Research: Smoking & Haemoglobin

Haemoglobin decreased (AUC 0.890, F1 0.522) appeared among the 50 tracked AEs. Smoking is a known pharmacokinetic disruptor — CYP1A2 induction from cigarette smoke accelerates metabolism of drugs like clozapine and theophylline, mimicking DDI effects. Incorporating smoking status as a feature in future work could improve haemoglobin-related AE detection.

Limitations & Future Directions

CYP450 Missingness

59.5% of Drug A and 47.1% of Drug B CYP450 profiles were missing. CYP overlap is a primary DDI predictor. Imputation or external CYP databases could recover this signal.

SMILES as Raw Text

SMILES strings encode molecular topology but TF-IDF tokenizes them as arbitrary characters. Molecular fingerprints (Morgan/ECFP) or graph neural networks would capture true structural similarity.

CV Subset Limitation

5-fold CV was run on 3,000 observations for runtime efficiency. This may not fully represent the 12,227-observation training distribution.

Label Independence

The MultiOutput classifier treats all 50 AE labels independently, ignoring co-occurrence structure (e.g., Nausea and Vomiting are highly correlated). Classifier chains could leverage this structure.

Cold-Start Test Set

The 3,974 held-out pairs have new molecular scaffolds not seen in training. Text-only models are structurally disadvantaged — graph-based representations would generalize better to unseen chemistry.

Future Directions

- Molecular fingerprints (Morgan/ECFP4) for PRR and cold-start generalization
- CYP450 overlap features from external databases (e.g., DrugBank)
- Classifier chains for correlated AE label prediction
- SMOTE or class-weight tuning for Minor class recall
- Graph neural network for drug pair structural encoding
- Smoking / lifestyle covariates for AE modulation modeling

Conclusions

01

Text features carry real clinical signal

TF-IDF on FDA FAERS pharmacological text achieved AUC > 0.91 for severity and AUC > 0.89 mean for adverse event prediction — strong performance on an unseen cold-start scaffold set.

02

Gradient Boosting outperformed all baselines

GB achieved the highest Macro F1 (0.5666) and AUC (0.9252). The narrow gap between models confirms the ceiling is features, not algorithms.

03

Class imbalance remains the dominant challenge

Minor class recall peaked at 19.8%. Addressing the 16.8:1 imbalance through SMOTE, class weighting, or threshold tuning is the most direct path to improvement.

04

AE clusters are severity-independent (RQ2)

K-means found high-burden vs. low-burden drug pair profiles, but severity composition was near-identical. Adverse event burden and severity are independent dimensions.

05

PRR requires structural features

Mean inverse RMSE of 0.2023 confirms TF-IDF is insufficient for continuous pharmacovigilance signal. Molecular fingerprints are the next step.

Anticipated Questions & Answers

Q: Why use Macro F1 instead of accuracy for severity?

A: Accuracy is misleading with a 16.8:1 imbalance — predicting "Moderate" for everything yields 69.1% accuracy with zero clinical value. Macro F1 weights all three classes equally, so the model is penalized for ignoring Minor and Major cases.

Q: Why was PRR so much weaker than the other tasks?

A: Three reasons: (1) only 21.5% of PRR entries are non-zero, making the signal sparse; (2) TF-IDF on SMILES lacks structural resolution for continuous regression; (3) high-variance events like Atrial Fibrillation have PRR values that are hard to predict from text alone.

Q: Why didn't you use SMOTE or class weighting to fix the imbalance?

A: These are valid next steps listed in future work. The priority was establishing a clean baseline pipeline. Class weighting in GB is a direct and low-risk next step that would likely improve Minor recall substantially.

Q: Why did the Kaggle score (0.5134) differ from your estimated composite (0.4605)?

A: The estimated composite was computed using my internal held-out test set. The Kaggle evaluation used a different scoring methodology and the actual competition test set. The divergence suggests my internal evaluation may have been conservative.

Q: How were the clusters determined?

A: K-means was applied to the 50 binary adverse event columns only. K was chosen by sweeping k=2 through k=6 and selecting the value with the highest silhouette score. k=2 scored 0.776 — higher cohesion and separation than any k from 3 to 6, which declined monotonically.

Q: What would be the highest-impact improvement?

A: Molecular fingerprints (e.g., Morgan/ECFP4 from RDKit). They encode true structural similarity and would improve both PRR regression and cold-start generalization to unseen scaffolds. Combined with CYP450 imputation from DrugBank, these two additions would likely lift all three task scores.